

Lecture Note: Representation of the “Tower” Network

1. Goal

We want a neural network that outputs

$$y = 1$$

only when the input point

$$x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}$$

lies inside the square

$$-0.5 < x_0 < 0.5, \quad -0.5 < x_1 < 0.5,$$

and outputs

$$y = 0$$

otherwise.

If we plot the output vertically over the (x_0, x_1) -plane, this creates a rectangular **tower**.

2. Step Activation Function

We use the threshold activation

$$\theta(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

applied elementwise to vectors.

3. Dimension-Consistent Network

A clean version of the network is a $\mathbf{2} \rightarrow \mathbf{4} \rightarrow \mathbf{2} \rightarrow \mathbf{1}$ threshold network.

Layer 1: Four wall tests

Let

$$W^{(1)} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}.$$

Then

$$z^{(1)} = W^{(1)}x + b^{(1)} = \begin{bmatrix} x_0 + 0.5 \\ -x_0 + 0.5 \\ x_1 + 0.5 \\ -x_1 + 0.5 \end{bmatrix}, \quad a^{(1)} = \theta(z^{(1)}).$$

So the four neurons test:

- $a_1 = \theta(x_0 + 0.5)$: checks $x_0 > -0.5$
- $a_2 = \theta(-x_0 + 0.5)$: checks $x_0 < 0.5$
- $a_3 = \theta(x_1 + 0.5)$: checks $x_1 > -0.5$
- $a_4 = \theta(-x_1 + 0.5)$: checks $x_1 < 0.5$

Thus, Layer 1 creates the four boundaries of the square.

Layer 2: Two interval tests

Now combine the four wall tests into two “inside-interval” tests:

$$W^{(2)} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} -1.5 \\ -1.5 \end{bmatrix}.$$

Then

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)} = \begin{bmatrix} a_1 + a_2 - 1.5 \\ a_3 + a_4 - 1.5 \end{bmatrix}, \quad a^{(2)} = \theta(z^{(2)}).$$

Write

$$a^{(2)} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

Then:

- $b_1 = \theta(a_1 + a_2 - 1.5)$ is 1 only if both a_1 and a_2 are 1.
- $b_2 = \theta(a_3 + a_4 - 1.5)$ is 1 only if both a_3 and a_4 are 1.

This means:

- $b_1 = 1$ iff $-0.5 < x_0 < 0.5$
- $b_2 = 1$ iff $-0.5 < x_1 < 0.5$

Why is -1.5 used?

Because for binary inputs $u, v \in \{0, 1\}$,

$$\theta(u + v - 1.5) = 1 \iff u = v = 1.$$

So **-1.5 implements a 2-out-of-2 threshold**, which is exactly an AND gate for two binary inputs.

Layer 3: Final inside-square test

Now require both coordinate interval tests to be true:

$$W^{(3)} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad b^{(3)} = -1.5.$$

Then

$$z^{(3)} = W^{(3)}a^{(2)} + b^{(3)} = b_1 + b_2 - 1.5, \quad y = \theta(z^{(3)}).$$

Therefore,

$$y = 1 \iff b_1 = b_2 = 1.$$

which means

$$y = 1 \iff -0.5 < x_0 < 0.5 \text{ and } -0.5 < x_1 < 0.5.$$

This is the corrected mathematical representation of the network.

4. Full Network in Compact Form

$$a^{(1)} = \theta(W^{(1)}x + b^{(1)}),$$

$$a^{(2)} = \theta(W^{(2)}a^{(1)} + b^{(2)}),$$

$$y = \theta(W^{(3)}a^{(2)} + b^{(3)}).$$

with

$$W^{(1)} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}.$$

$$W^{(2)} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} -1.5 \\ -1.5 \end{bmatrix}.$$

$$W^{(3)} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad b^{(3)} = -1.5.$$

5. Layer-by-Layer Interpretation

Layer	Role	Mathematical meaning
Layer 1	Four wall detectors	Tests which side of each boundary line the point lies on
Layer 2	Two interval detectors	Checks whether x_0 and x_1 are each within their bounds
Layer 3	Final AND	Declares whether the point is inside the square

So the network works as:

$$(x_0, x_1) \rightarrow 4 \text{ wall tests} \rightarrow 2 \text{ interval tests} \rightarrow \text{inside-square output}.$$

6. Worked Example 1: Point Inside the Square

Take

$$x = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}.$$

Layer 1

$$z^{(1)} = W^{(1)}x + b^{(1)} = \begin{bmatrix} 0.2 + 0.5 \\ -0.2 + 0.5 \\ 0.1 + 0.5 \\ -0.1 + 0.5 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.3 \\ 0.6 \\ 0.4 \end{bmatrix}.$$

so

$$a^{(1)} = \theta(z^{(1)}) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

Layer 2

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1.5 \\ -1.5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}.$$

so

$$a^{(2)} = \theta(z^{(2)}) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Layer 3

$$z^{(3)} = W^{(3)}a^{(2)} + b^{(3)} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 1.5 = 0.5.$$

hence

$$y = \theta(z^{(3)}) = 1.$$

So the point is **inside** the square.

7. Worked Example 2: Point Outside the Square

Take

$$x = \begin{bmatrix} 0.8 \\ 0.1 \end{bmatrix}.$$

Layer 1

$$z^{(1)} = W^{(1)}x + b^{(1)} = \begin{bmatrix} 0.8 + 0.5 \\ -0.8 + 0.5 \\ 0.1 + 0.5 \\ -0.1 + 0.5 \end{bmatrix} = \begin{bmatrix} 1.3 \\ -0.3 \\ 0.6 \\ 0.4 \end{bmatrix}.$$

so

$$a^{(1)} = \theta(z^{(1)}) = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

so

$$a^{(2)} = \theta(z^{(2)}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Layer 2

$$z^{(3)} = W^{(3)}a^{(2)} + b^{(3)} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 1.5 = -0.5.$$

hence

$$y = \theta(z^{(3)}) = 0.$$

so

$$a^{(2)} = \theta(z^{(2)}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Layer 3

$$z^{(3)} = W^{(3)}a^{(2)} + b^{(3)} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 1.5 = -0.5.$$

hence

$$y = \theta(z^{(3)}) = 0.$$

So the point is **outside** the square.

8. Summary Table

Point	Layer 1 output $a^{(1)}$	Layer 2 output $a^{(2)}$	Final output y
$\begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	1
$\begin{bmatrix} 0.8 \\ 0.1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	0

9. Main Takeaway

This representation shows how layers in a threshold network can be interpreted geometrically and logically:

1. **Layer 1** creates half-space tests.
2. **Layer 2** combines pairs of tests into interval checks using the threshold **-1.5**.
3. **Layer 3** combines the two interval checks into a final inside-square decision.

Thus, multilayer networks can build localized regions in input space, which is the basic idea behind constructing “towers” and, more generally, approximating complicated functions.